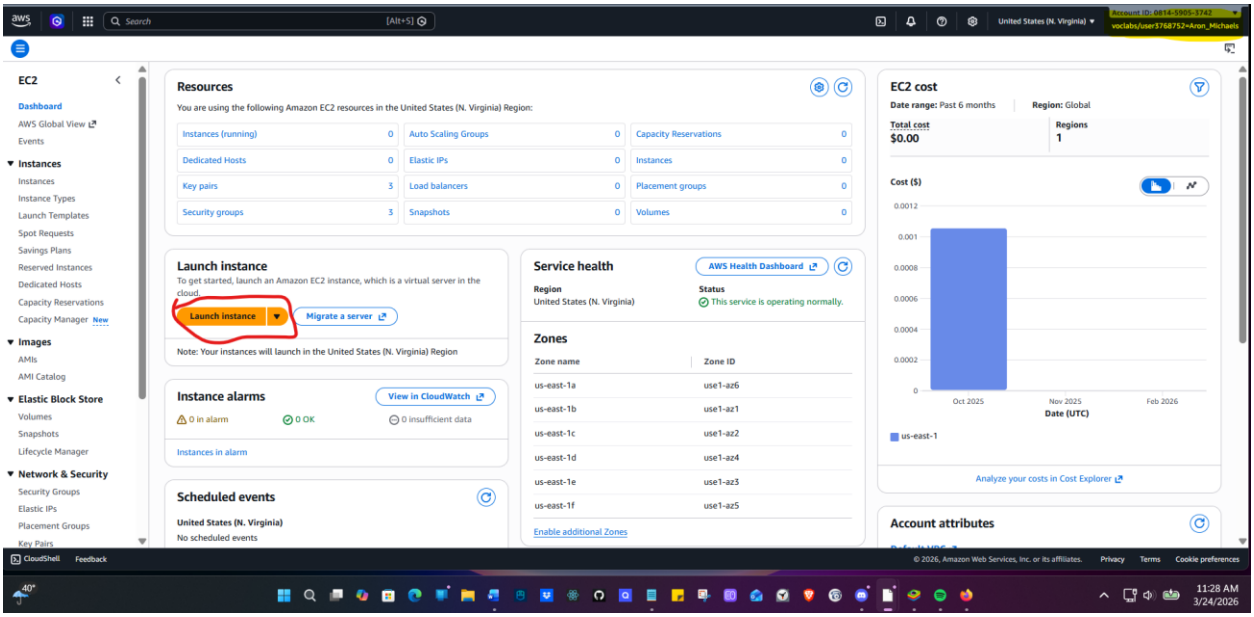
Week 2 Project Template **Aron Michaels**

CMIT 495 Current Trends and Projects in Computer Networks and Security

Week 2 – Cloud Computing

1. **Access your AWS Learner Lab and take a screenshot of the AWS Management Console (Dashboard) and paste it below question 1. The screenshot should include the username you were assigned during the setup phase.**

I accessed the AWS Learner Lab and opened the Management Console dashboard in the us-east-1 region. The screenshot below shows me assigned username and confirms the lab environment is ready for the Caduceus Inc. cloud project.



Dashboard

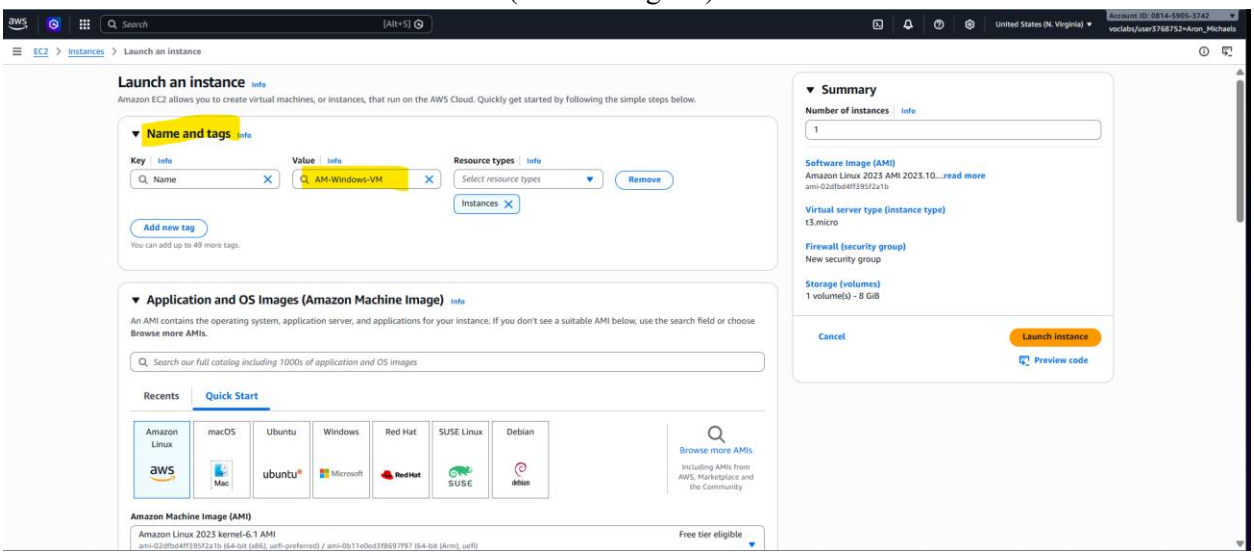
2. **Launch a Windows Virtual Machine (VM). Provide a detailed overview of the steps required to install the Microsoft Windows operating system (OS) on the VM. The steps may be listed in the form of bullet points or a summary with complete sentences. Use as much space as required. Finally, take a screenshot of the desktop and paste it with your response below this question.**

I provisioned a Microsoft Windows Server

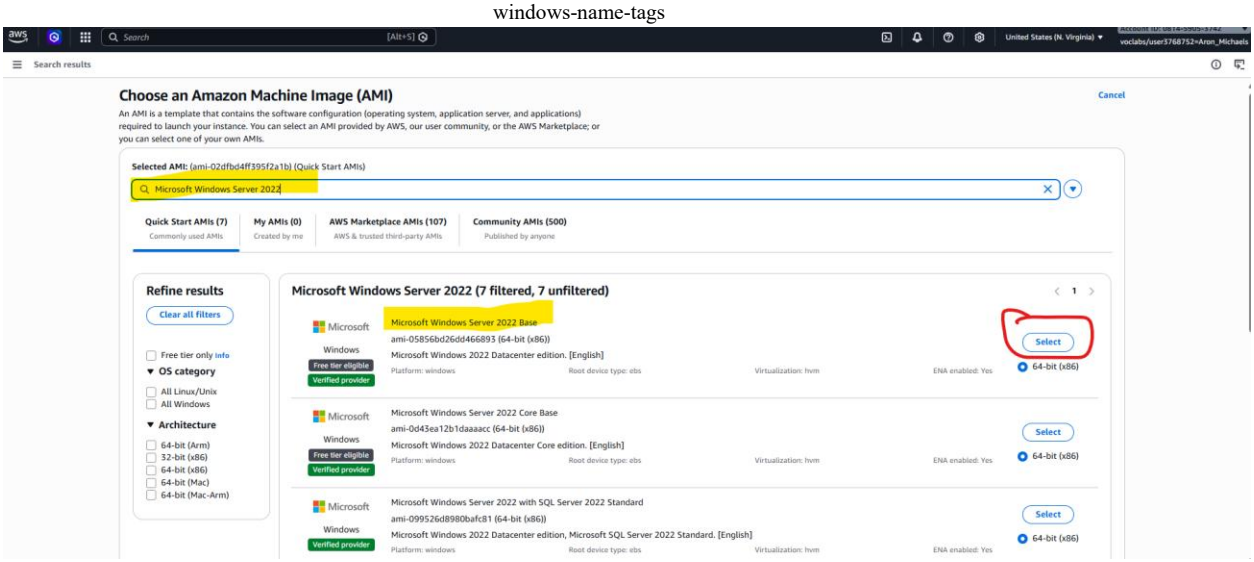
2022 virtual machine using the AWS EC2 console with a PaaS approach.

Steps:

- Selected Windows Server 2022 Base AMI (free tier eligible).

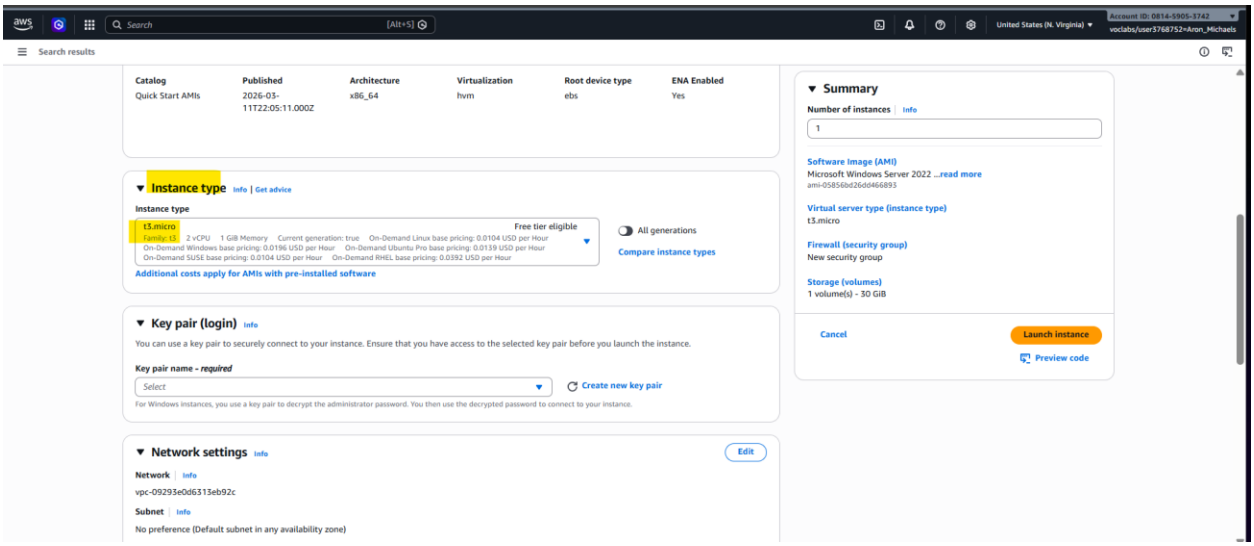


windows-name-tags



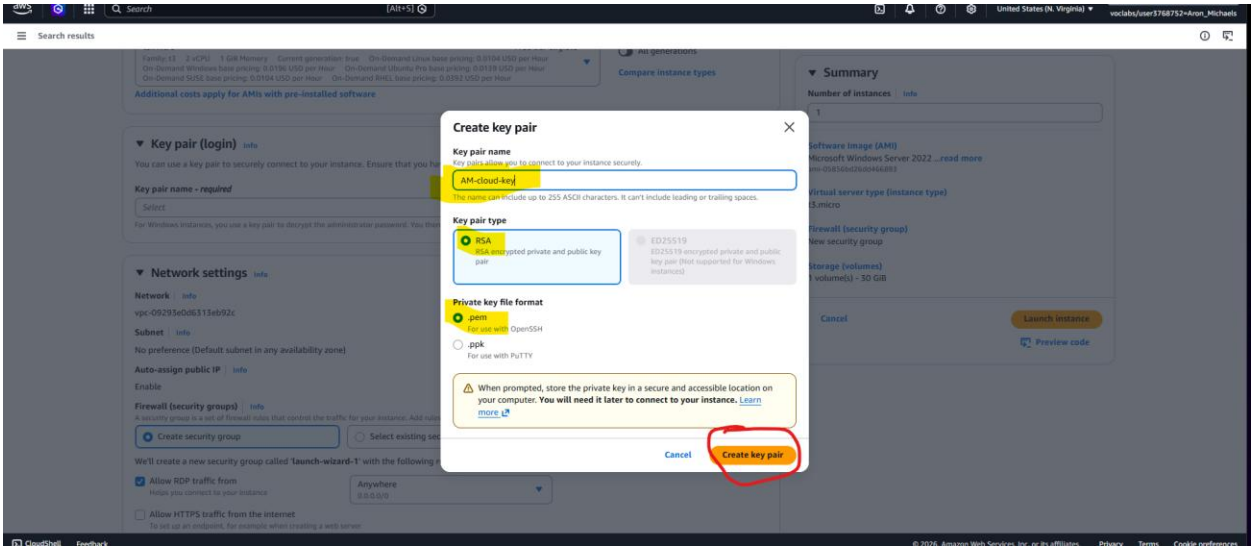
windows-ami

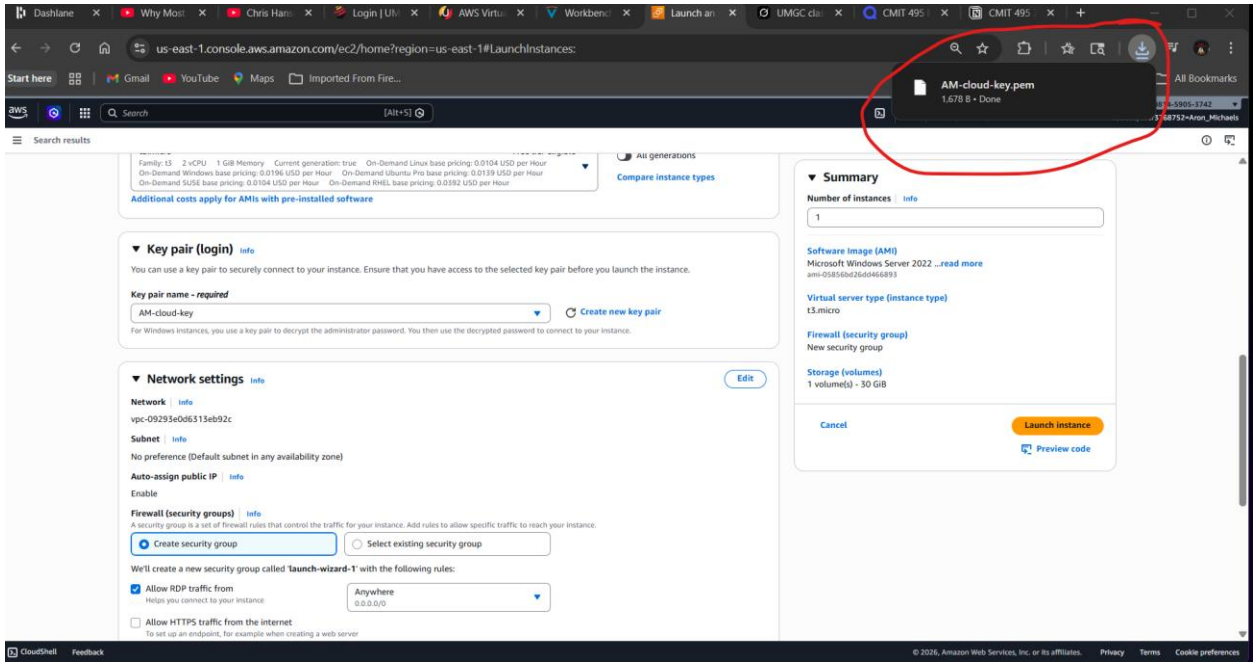
- Chose t3.micro instance type



windows-type

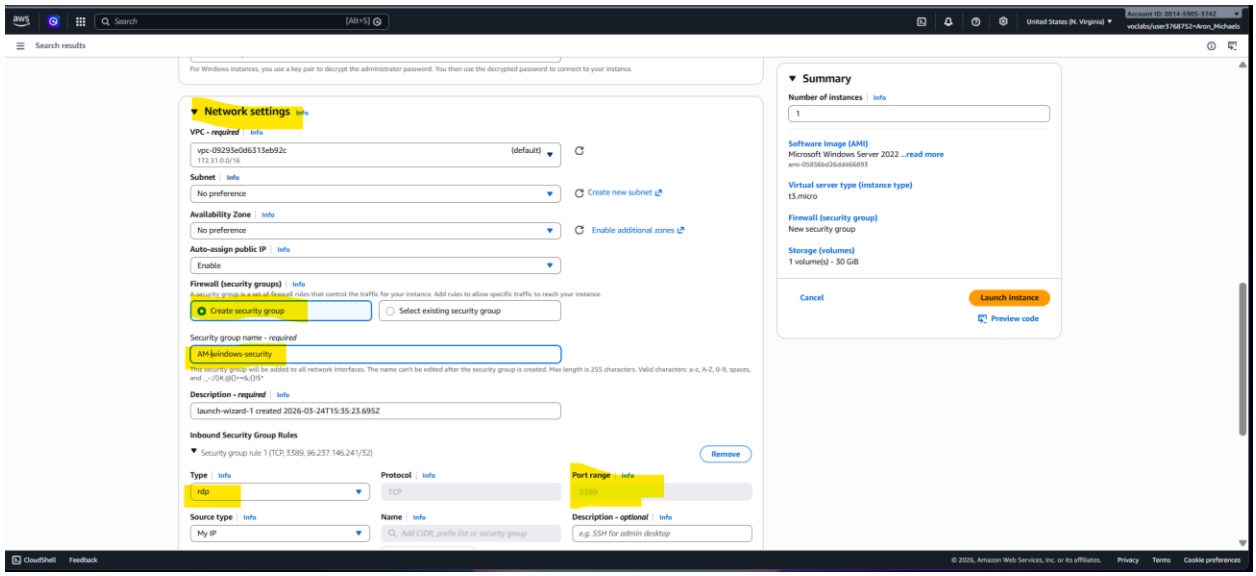
- Created new key pair (grace-cloud-key.pem).



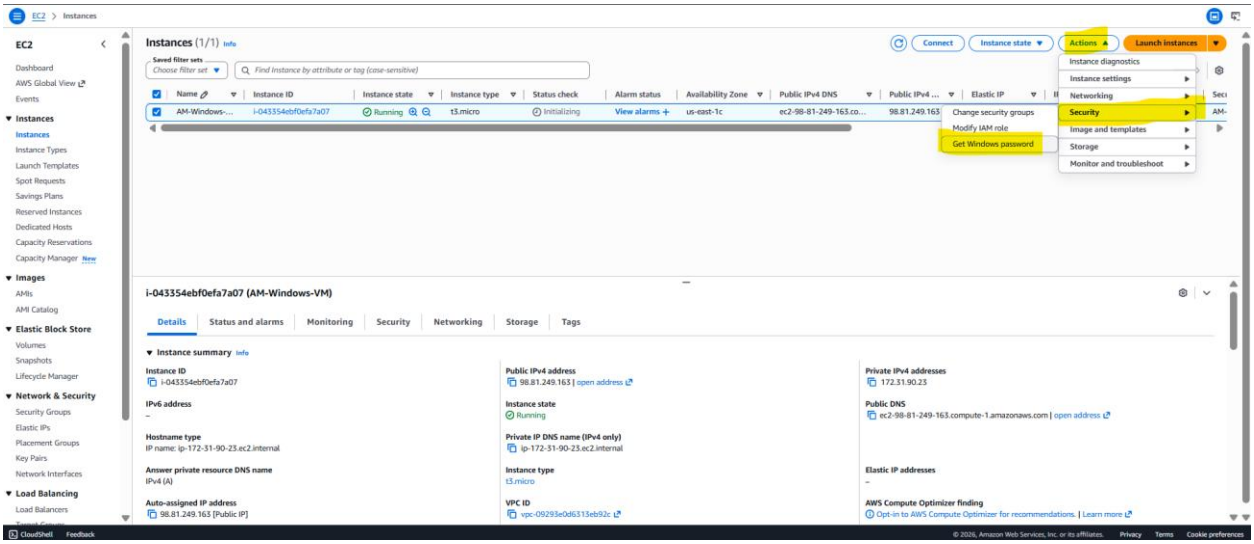


Windows key pari download

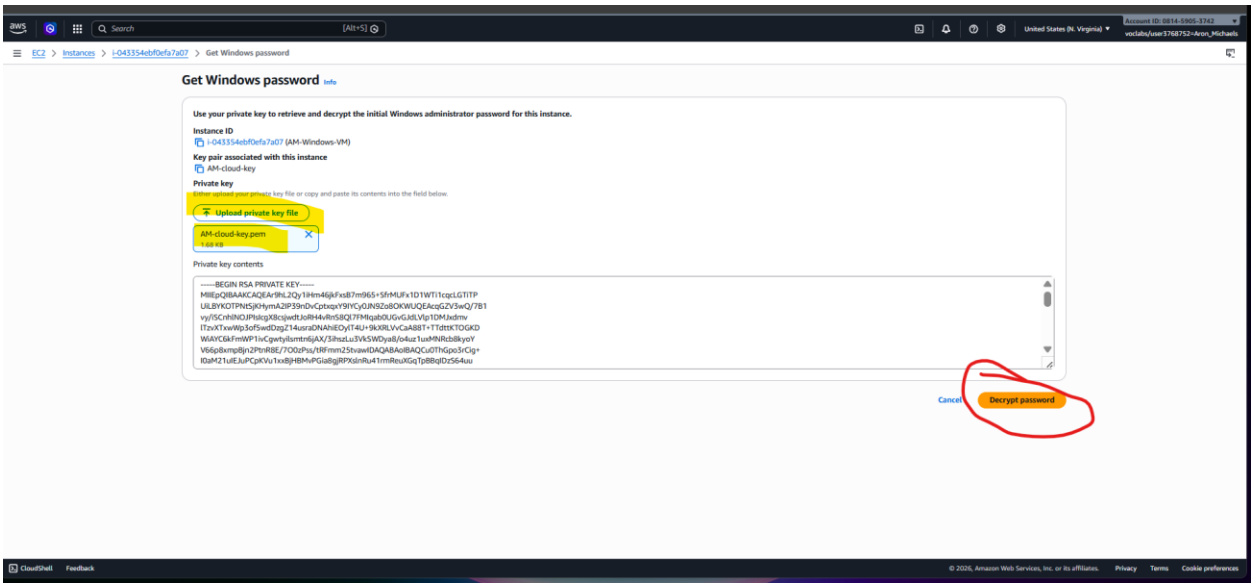
- Configured security group for RDP access from My IP only.



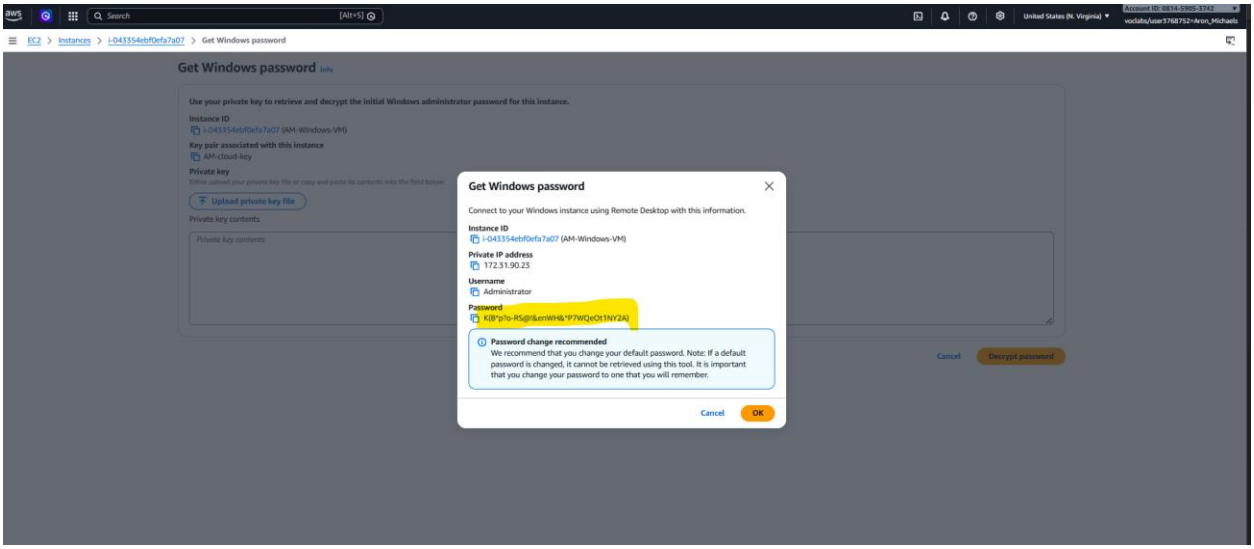
- Launched the instance and connected via Remote Desktop Connection using the decrypted Administrator password.



Get Windows Password

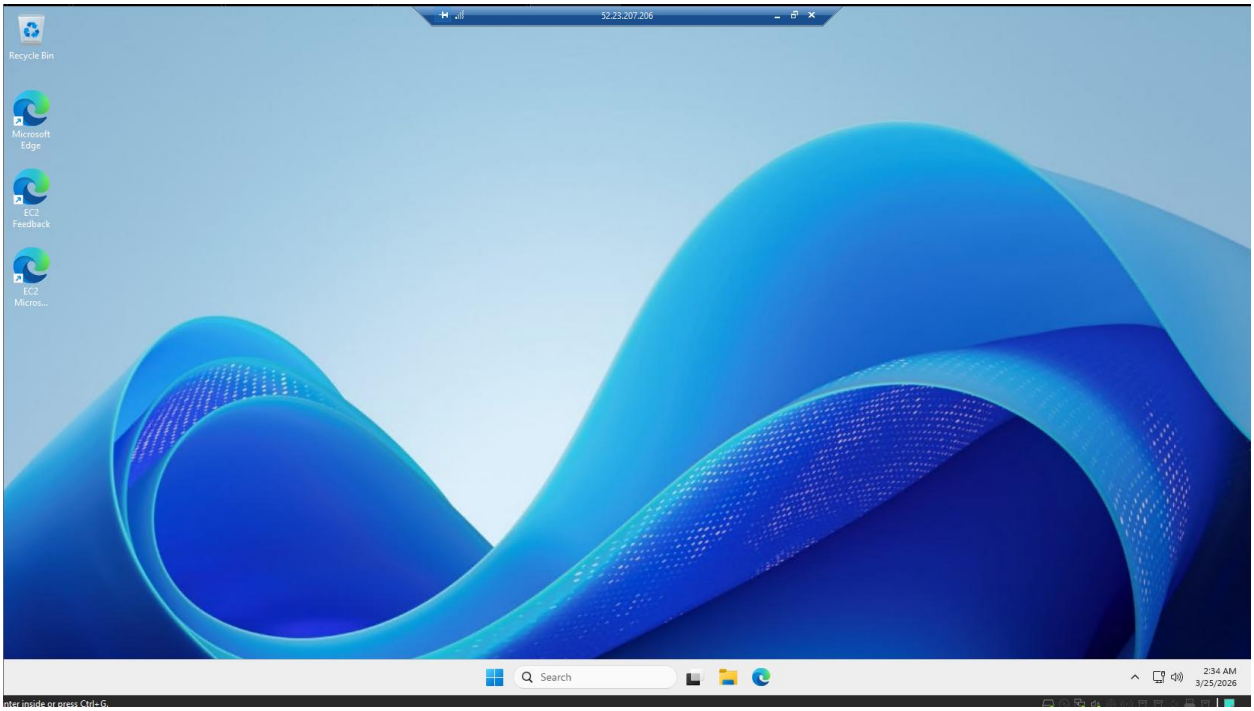


Decrypt windows password



Windows password shown

- The final screenshot shows the successful Windows desktop, proving the OS is installed and accessible from any end-user device as required by CTO Eliza.



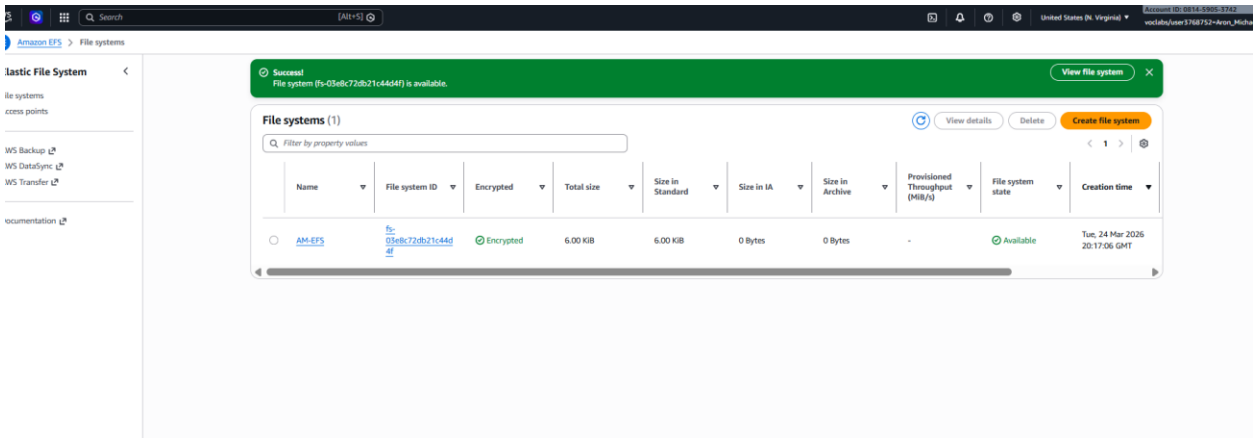
Windows server desktop VM

3. **Using what you learned from Project 1, provision and launch a new AWS EC2 Ubuntu Linux Server and connect to it via the SSH protocol. Note any challenges or opportunities associated with this provisioning.**

   Using knowledge from Project 1, I launched a

new AWS EC2 Ubuntu Server 24.04 LTS instance (t3.micro) and connected via SSH with
PuTTY. No new challenges occurred because the steps were identical to the previous project.
Opportunity: This demonstrates how easily we can provision both Windows and Linux in the
same AWS environment for Caduceus Inc. team collaboration.

4. **Using AWS, *create a network file system* with Amazon Elastic File Systems (EFS) and attach it to the running Ubuntu Server instance. You may use the <u>AWS</u> *web page* for step-by-step instructions and understand how the EFS works. Take a screenshot of the result and embed it below. Specifically, take a screenshot to verify *that your file system has been successfully mounted*, along with the results from *creating a test file in your new file system. This will be done by running a simple dd command to generate a 1GiB file in your new directory.* Finally, describe the value of a network file system.**



efs-created

```
ubuntu@ip-172-31-92-236:~$ df -h
Filesystem                                          Size  Used Avail Use% Mounte
d on
/dev/root                                           6.8G  2.1G  4.7G  31% /
tmpfs                                               456M     0  456M   0% /dev/s
hm
tmpfs                                               183M  892K  182M   1% /run
tmpfs                                               5.0M     0  5.0M   0% /run/l
ock
efivarfs                                            128K  3.6K  120K   3% /sys/f
irmware/efi/efivars
/dev/nvme0n1p16                                     881M   94M  726M  12% /boot
/dev/nvme0n1p15                                     105M  6.2M   99M   6% /boot/
efi
tmpfs                                                92M   12K   92M   1% /run/u
ser/1000
fs-0bf08caf60b2dfa8d.efs.us-east-1.amazonaws.com:/  8.0E     0  8.0E   0% /mnt/e
fs
ubuntu@ip-172-31-92-236:~$
```

efs-mounted-1gib

I created an Amazon Elastic File System (EFS) and attached it to the running Ubuntu Server instance using the NFS client as recommended in the EFS Hints document.

Steps performed:

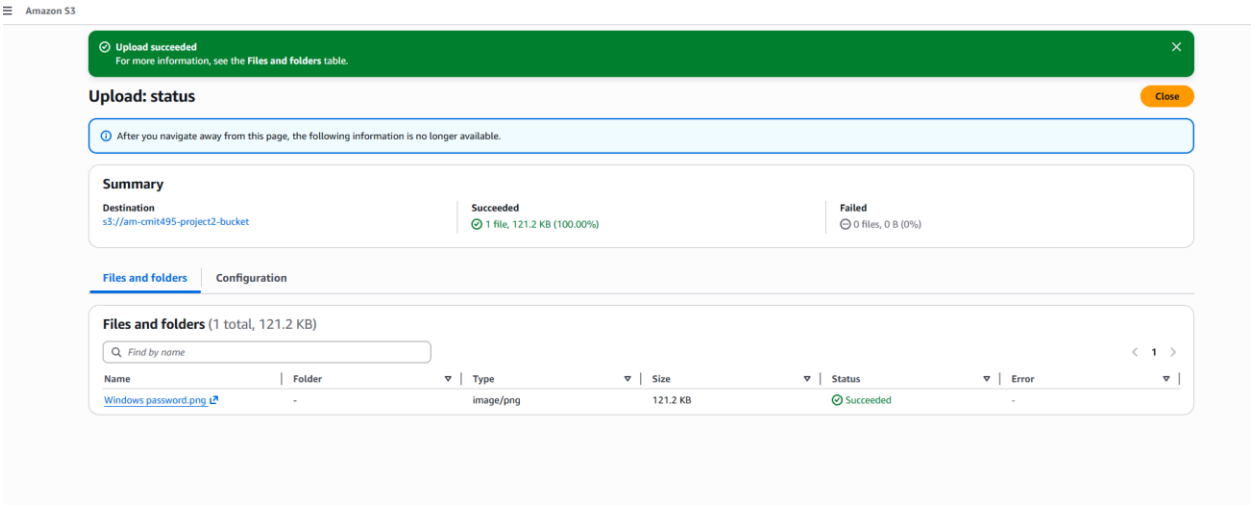
- Created the EFS file system in the AWS console.
- Installed nfs-common on the Ubuntu instance.
- Mounted the EFS using the command: sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport [my-fs-dns].efs.us-east-1.amazonaws.com:/ /mnt/efs
- Verified the mount with df -h.
- Created a 1GiB test file using dd and confirmed it with ls.

The screenshot above shows the EFS successfully mounted and the 1GiB test file present in the shared directory.

Value of a network file system: A network file system like EFS allows multiple virtual machines (such as Windows and Linux instances) to share the same storage space

in real time. This is very useful for Caduceus Inc. because team members working on different operating systems can access and update the same files without duplication, improving collaboration and data consistency while supporting the company's telehealth operations.

5. **Using the AWS platform, create an S3 bucket and upload any file to the S3 bucket. Take a screenshot showing the file was uploaded to the S3 bucket and paste it below. If necessary, use the AWS webpage above for step-by-step instructions.**



s3-upload

I created an S3 bucket named grace-cmit495-project2-bucket and uploaded a test file to it. The screenshot above shows the file successfully stored in the S3 bucket.

6. **The CTO will be reviewing this document. You have shown how easy it is to provision a Microsoft OS using the AWS platform. The CTO chose AWS because it offered a free account. She will now expect a recommendation from you on what cloud service to use for the organization's PaaS (e.g., the infrastructure, OS, runtime, etc.) needs. There is no need for a private cloud, so the public option will work just fine. Describe the difference between the Google Cloud platform, Amazon AWS platform, and Microsoft Azure platform. Provide a**

recommendation for the CTO as to which service provider you would recommend and why. Be explicit and detailed in your recommendation.

The CTO asked for a recommendation on which public cloud provider to use for Caduceus Inc.'s PaaS needs (infrastructure, operating systems, and runtime).

Amazon AWS, Google Cloud Platform, and Microsoft Azure are the three major public cloud providers.

Amazon AWS offers a generous free tier, excellent support for both Windows and Linux VMs, and native Elastic File System (EFS) for shared storage, which made this project very straightforward.

Google Cloud Platform is strong in data analytics and machine learning but has a steeper learning curve for beginners and less seamless Windows integration.

Microsoft Azure provides very good Windows integration and hybrid cloud options, but its free tier is smaller and shared storage setup (Azure Files) is more complex than AWS EFS.

I recommend Amazon AWS for Caduceus Inc. It matches the free account the CTO already likes, provides simple PaaS for both Windows and Linux, includes easy shared storage with EFS, and follows NIST cloud characteristics such as rapid elasticity and measured service. This will allow the company to scale efficiently while keeping costs low and management simple.

7. **The CTO approved your comparative analysis between the cloud service providers (i.e. Amazon, Google, and Microsoft). She has decided to proceed with an Amazon Virtual Private Cloud (Amazon VPC). The Amazon VPC enables one to launch AWS resources into a virtual network, which is similar to a traditional network that can be operated in an on-premises data**

center. Keep in mind that networking, storage, and security associated with a VPC are as important as the overall scalable infrastructure of AWS.

a. **To begin, the CEO would like you to provide the network settings needed to provision two (2) subnets for the VPC as shown in the table below:**

Subnet	End-User	CIDR	Network	Broadcast	Mask
A	**Developers**	**146.38.70.105/20**	*Click or tap here to enter text.*	*Click or tap here to enter text.*	*Click or tap here to enter text.*
B	**Marketing**	**172.31.0.0 /16**	*Click or tap here to enter text.*	*Click or tap here to enter text.*	*Click or tap here to enter text.*

b. **Based on your understanding, list the network address, broadcast address, and subnet mask for subnet A and subnet B in the table above. Perform the necessary calculations and explain how you arrived at your answer.**

To better help you understand IP addressing, IP subnetting, and IP address summarization, review the following AWS documentation prior to answering the questions in this section:

- **VPCs and subnets**
- **CIDR and Peering for VPC and AWS Control Tower**
- **Subnet CIDR reservations**

Note:

The key benefit of an Amazon VPC (or a virtual private network) is that the internal network devices are not openly accessible via the Internet and can only be accessed from within a secure network. Thus, it keeps the proprietary applications and data protected.

Classless Internet Domain Routing (CIDR) notation: CIDR was introduced as a means to primarily improve address space utilization as a result of the rapid growth of the Internet and growth of the IP routing tables held in the Internet routers. Represented by an IP prefix, CIDR moves away from the traditional IP classes (e.g., Class A, Class B, Class C, etc.). Subnetting a network address space using CCIDR leads to an effective IP address space only for the number of hosts needed without wasting IP addresses.

For Subnet A (146.38.70.105/20): The /20 prefix means the subnet mask is 255.255.240.0. Applying the mask to the IP gives the network address 146.38.64.0. The broadcast address is the last address in that range: 146.38.79.255. For Subnet B (172.31.0.0/16): The /16 prefix gives the subnet mask 255.255.0.0. The network address is 172.31.0.0 and the broadcast address is 172.31.255.255.

These calculations follow standard CIDR subnetting rules from the AWS VPC documentation. The Amazon VPC keeps internal resources secure by not exposing them directly to the internet.

8. **Please note the following carefully. Confirm that you have <u>stopped</u> and <u>terminated</u> your Microsoft Windows virtual machine, <u>deleted</u> your file system from the Amazon EFS console, <u>deleted</u> the contents of your Amazon S3 bucket, and <u>deleted</u> your Amazon S3 bucket. To confirm, simply type your name below.**

   Aron Michaels

Export/print your completed file as a PDF and upload to your LEO/Assignments folder.